A particle P with constant speed v is moving around a fixed circle with radius r. Find its acceleration.

Let's solve this **step by step**.

We are asked: a particle P moves with **constant speed** v along a **circle of radius** r. We need to find its **acceleration**.

Step 1: Understand the motion

- The particle is moving in a **circle** at **constant speed** → this is **uniform circular motion**.
- Even though the **speed is constant**, the **velocity vector changes direction continuously**.
- Acceleration is related to the **change in velocity**, so there **is acceleration** directed toward the **center of the circle**. This is called **centripetal acceleration**.

Step 2: Magnitude of acceleration

For uniform circular motion:

$$a = \frac{v^2}{r}$$

Where:

- v = constant speed of the particle
- r = radius of the circle
- a = magnitude of **centripetal acceleration**

Step 3: Direction of acceleration

- The acceleration is **always directed toward the center of the circle** (radially inward).

✅ Step 4: Conclusion

- **Magnitude:** $a = \frac{v^2}{r}$
- **Direction:** Toward the center of the circle (radial/centripetal).

If you want, I can also **derive it using vector calculus** to show exactly why $a = v^2/r$ using $\vec{r}(t)$ and derivatives—it's a neat demonstration.

Do you want me to do that?

yes

Great! Let's derive it **rigorously using vector calculus**. We'll go step by step.

Step 1: Represent the position vector

Let's put the circle in the xy-plane, with **center at the origin**.
If the particle moves **counterclockwise**, its position vector at time t is:

$$\vec{r}(t) = r\cos\theta(t)\,\hat{i} + r\sin\theta(t)\,\hat{j}$$

Where:

- r = radius of the circle
- $\theta(t)$ = angular position of the particle at time t
- $\hat{i}, \hat{j}$ = unit vectors along x and y

Step 2: Relate angular speed and linear speed

The **speed** v is constant:

$$v = r\frac{d\theta}{dt} \quad \Rightarrow \quad \frac{d\theta}{dt} = \frac{v}{r}$$

Call this constant ω, the **angular speed**:

$$\omega = \frac{v}{r}$$

So $\theta(t) = \omega t = \frac{v}{r}t$.

Step 3: Velocity vector

Velocity is the **time derivative of position**:

$$\vec{v}(t) = \frac{d\vec{r}}{dt} = \frac{d}{dt}\left(r\cos\theta\,\hat{i} + r\sin\theta\,\hat{j}\right)$$

Use the chain rule:

$$\vec{v}(t) = -r\sin\theta\,\frac{d\theta}{dt}\,\hat{i} + r\cos\theta\,\frac{d\theta}{dt}\,\hat{j}$$

Substitute $\frac{d\theta}{dt} = \omega$:

$$\vec{v}(t) = -r\omega\sin\theta\,\hat{i} + r\omega\cos\theta\,\hat{j}$$

- Magnitude:

$$|\vec{v}| = r\omega = v$$

✅ Perfect, the speed is constant.

Step 4: Acceleration vector

Acceleration is the **time derivative of velocity**:

$$\vec{a}(t) = \frac{d\vec{v}}{dt} = \frac{d}{dt}\Big(-r\omega\sin\theta\,\hat{i} + r\omega\cos\theta\,\hat{j}\Big)$$

Again, use the chain rule ($d\theta/dt = \omega$):

$$\vec{a}(t) = -r\omega^2\cos\theta\,\hat{i} - r\omega^2\sin\theta\,\hat{j}$$

Factor out $-\omega^2$:

$$\vec{a}(t) = -\omega^2(r\cos\theta\,\hat{i} + r\sin\theta\,\hat{j}) = -\omega^2\vec{r}(t)$$

Step 5: Magnitude of acceleration

$$|\vec{a}| = \omega^2 r$$

Substitute $\omega = \frac{v}{r}$:

$$|\vec{a}| = \left(\frac{v}{r}\right)^2 r = \frac{v^2}{r}$$

✅ Step 6: Direction

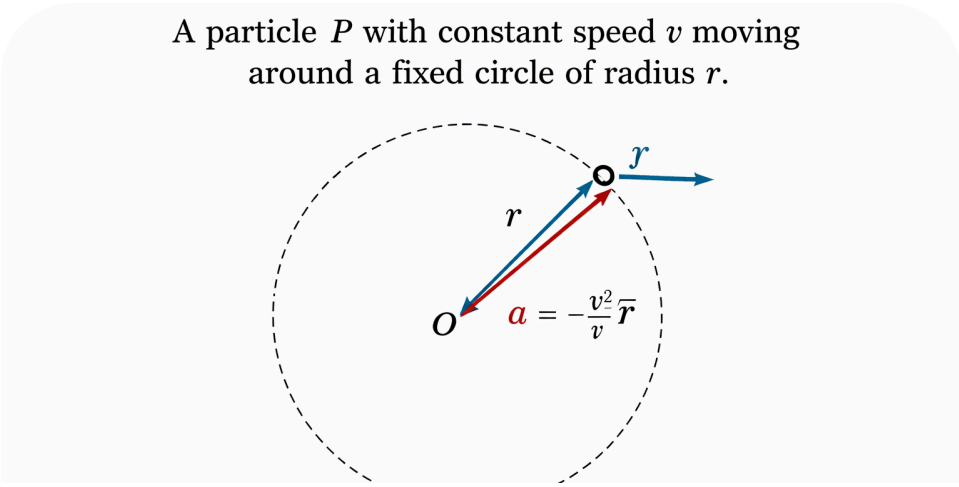
$$\vec{a} = -\omega^2\vec{r}$$

- The acceleration vector points **opposite to $\vec{r}$ → toward the center of the circle**.
- This is exactly the **centripetal acceleration**.

✅ Conclusion

- **Acceleration magnitude:** $a = \frac{v^2}{r}$
- **Direction:** Radially inward toward the center.

If you want, I can also **draw a diagram showing $\vec{r}, \vec{v}, \vec{a}$ vectors for clarity**, which makes this very intuitive.

Do you want me to make that diagram?

yes